

TASE-Net: A Novel Robust Deep-Learning Network for Open-Set Few-Shot UAV Recognition

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Abstract—As uncrewed aerial vehicles (UAVs) are employed for many applications, UAV identification becomes critical to air traffic control nowadays. Conventional radio-frequency (RF) based UAV identification schemes can correctly recognize a UAV according to the acquired sufficient training RF signal data from a prespecified training candidate set. However, they may often fail when the model of a test UAV is not included in such a training candidate set and/or the training data are quite limited. To address the aforementioned practical challenges, a novel RF-based open-set few-shot UAV recognition technique is introduced in this work. In our proposed new approach, an RF signal of interest, such as a UAV control signal and a digital data-transmission (DDT) signal, is first sensed and segmented in the UAV frequency band using the corresponding short-time energy-to-spectral entropy ratio (ST-ESER). Then, the cyclic spectrum analysis is applied to the selected RF signal segments to construct the corresponding polyspectra, which are further utilized to establish the cyclic-paw-print (CPP) tensors. Moreover, we design a novel deep-learning (DL) network, namely, transformer-attention squeeze-excitation network (TASE-Net), by fusing the transformer-enhanced squeeze-and-excitation (SE) model and the Gaussian mixture model (GMM) into the residual network. The TASE-Net can excel in global feature modeling and unknown class detection simultaneously especially for the open-set and few-shot scenarios. Finally, the constructed CPPs are adopted as the input features of our proposed new TASE-Net to recognize the RF signals (the UAV models). Monte Carlo simulation results demonstrate that our proposed new UAV recognition approach using the TASE-Net can greatly outperform other existing DL methods for open-set few-shot UAV identification.

Index Terms—Cyclic-paw-print (CPP), open-set few-shot deep-learning RF recognition, short-time energy-to-spectral-entropy ratio (ST-ESER), transformer-attention squeeze-excitation network (TASE-Net), uncrewed aerial vehicle (UAV) identification.

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I. INTRODUCTION

THE EXPLOSIVE applications of uncrewed aerial vehicles (UAVs) in a wide variety of fields can be found in our daily life [1], [2], [3], [4], [5]. Meanwhile, the unauthorized use of UAVs also imposes serious threats on personal privacy, public safety, and aviation security [6]. It is quite challenging to identify the type of a UAV as it is a low-altitude, slow-speed, and small-size flying object. Hence, robust UAV detection and identification have been appealing to researchers recently. Numerous UAV identification methods using the radio-frequency (RF) features have been proposed [7]. The RF-based UAV sensing techniques, which can work effectively for both light-of-sight (LOS) and non-LOS (NLOS) scenarios, are deemed quite promising [8]. Furthermore, deep-learning (DL) models have been proposed (as classifiers) to carry out UAV sensing recently [9]. In particular, a lightweight backbone network consisting of lightweight multiscale convolution blocks was proposed to undertake UAV RF fingerprint identification [10]. A dynamic shrinkage learning network (DSLNN) was proposed to improve the accuracy of UAV RF fingerprint recognition for low signal-to-noise ratios (SNRs) and reduce the time complexity [11]. A fixed-boundary empirical wavelet feature-extraction strategy in tandem with a lightweight convolutional neural network (CNN) was proposed for UAV classification [12]. The minimum-variance distortionless-response (MVDR) spectrum of the received UAV RF signal was extracted and fed to a lightweight hybrid DL network, namely, UAV-CTNet, to carry out UAV detection and recognition in the presence of interference in our recent work [13]. Wavelet scattering transform (WST) was adopted to convert the received UAV control signals to images (scatter-plots) and then a CNN was utilized to identify UAV types thereupon [14]. To the best of our knowledge, almost all of the above-stated existing DL-based UAV sensing methods rely on the assumption that the UAV types to be recognized have to be included in a predefined UAV candidate set and a sufficiently large training dataset is available beforehand. As a matter of fact, we would often have new UAV type(s) out of the training dataset and the training samples of each UAV candidate model are often scarce in practice. The former challenge in the presence of unseen UAV type(s) is referred to as the “open-set problem” according to [15] while the latter challenge in the scarcity of training data is referred to as the “few-shot problem” according to [16].

Open-set pattern classification involves the decision to reject unknown (untrained) classes [17]. A DL mechanism, namely, “OpenMax layer,” which can estimate the probability of an

input being from an unknown (untrained) class, was proposed to carry out open-set image recognition [18], [19], [20]. Besides, an enhanced open-set recognizer was proposed to improve the recognition accuracy and the computational efficiency for open-set RF fingerprint recognition by establishing a proper open-set decision boundary through output calibration [21]. A meta supervised contrastive learning (MSCL) scheme, which can magnify interclass distinctions and reinforce intraclass compactness, was proposed to tackle few-shot open-set modulation classification using signal-constellation diagrams [22]. Since most of the existing open-set recognition techniques have been designed for image recognition, it would be interesting to extend the current open-set recognition approach to other applications such as UAV sensing. On the other hand, few-shot learning (FSL) has been investigated by researchers for decades due to its importance [23], [24], [25]. For example, an SSwsrNet framework using the deep residual shrinkage network (DRSN) and the semi-supervised learning paradigm was exploited for dynamic wireless communication-signal recognition given few training samples [26]. A hybrid DL model of a conditional generative adversarial network (CGAN) and a CNN was designed for few-shot jamming-signal recognition [27]. The neural architecture search (NAS) scheme and the advanced deep transfer learning (DTL) method were integrated to identify long-range (LoRa) devices through few-shot training [28]. The few-shot automatic modulation classification (AMC) was carried out using the Λ -DARTS algorithm [29]. Nonetheless, there hardly exists any few-shot UAV type identification technique up to now. We proposed a new robust signal feature, namely, cyclic-pawprint (CPP), which was extracted from the second-order cyclic spectrum (SCS) of the composite-modulation signal, for automatic composite-modulation classification (ACMC) lately [30], [31], [32]. In this work, a novel RF-based open-set few-shot UAV recognition scheme using CPP features is proposed while a new *transformer-attention squeeze-excitation network* (TASE-Net) is introduced here as the open-set UAV classifier. In our proposed new approach, a received UAV RF signal sequence is first decomposed into segments using uniformly spaced non-overlapping windows. Then, the corresponding “short-time energy-to-spectral-entropy ratio (ST-ESER)” of each segment can be calculated. Next, we only retain the segments having the k largest ST-ESERs among all. We call these k segments “*predominant segments*.” For each predominant segment, we create a CPP tensor, which will be utilized as the input of our proposed TASE-Net (the underlying open-set classifier). Our proposed new DL model, namely, TASE-Net, is designed by fusing the transformer-enhanced squeeze-and-excitation (SE) model and the Gaussian mixture model (GMM) into a residual network.

By using a publicly available dataset, the Monte Carlo simulations are conducted to demonstrate that our proposed new UAV recognition technique can not only reliably classify the pretrained UAV types but also properly reject the unseen (untrained) UAV types through FSL. Meanwhile, we also compare the performance of our proposed new scheme and those of three prevalent DL models for open-set few-shot UAV recognition to justify the effectiveness of our proposed

novel robust TASE-Net using CPP features. The primary contributions of this work can be highlighted as follows.

- 1) A novel segment-picking mechanism based on ST-ESER is proposed in our new approach to choose the predominant segments for robust UAV recognition.
- 2) The theoretical analysis about how many predominant segments need to be picked (retained) is facilitated using order statistics.
- 3) The robust CPP tensors are proposed to be the input features for UAV recognition.
- 4) A novel DL network, namely, TASE-Net, is designed by fusing the transformer-enhanced SE model and the GMM into a residual network for open-set few-shot UAV recognition.

The remainder of this article is organized as follows. Our proposed novel UAV identification approach is introduced in Section II. Monte Carlo simulations for the performance evaluation of our proposed new few-shot open-set UAV recognition technique are undertaken in Section III. Finally, conclusion will be drawn in Section IV.

Nomenclature: Scalars are denoted by italicized alphabets such as a ; vectors are denoted by alphabets with overhead arrow notions such as $\vec{A}$; matrices are denoted by alphabets with overhead tilde notions such as $\tilde{A}$; $\tilde{A}^T$ and $\tilde{A}^T$ represent the transposes of a vector $\vec{A}$ and a matrix $\tilde{A}$, respectively. Tensors are denoted by bold-faced alphabets such as $\mathbf{A}$. The set of all real numbers is denoted by $\mathbb{R}$ and the symbol $j \stackrel{\text{def}}{=} \sqrt{-1}$ is reserved to represent the unit imaginary number throughout this article. Note that $\otimes$ denotes the convolution between two tensors.

II. OUR PROPOSED NOVEL ROBUST UAV IDENTIFICATION APPROACH

The block diagram of our proposed novel open-set few-shot UAV identification technique is illustrated by Fig. 1, which consists of three major components: 1) RF signal acquisition and segmentation; 2) CPP extraction; and 3) TASE-Net. The received UAV RF signal sequence is first broken into uniformly spaced non-overlapping segments and the corresponding 3-D SCS is then constructed. Furthermore, the top view of the normalized SCS (NSCS) is converted to a CPP image, which can be represented by a CPP tensor. Finally, the TASE-Net, which is pretrained using the known UAV RF signals, will determine whether the received RF signal is unknown (untrained) and identify the type of the received UAV RF signal if it belongs to the pretrained candidate set. Now, let's assume that the pretrained UAV RF signal candidate set is denoted by $\mathbb{K} \stackrel{\text{def}}{=} \{\mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_K\}$. The details of the aforementioned three major components of our proposed new open-set few-shot UAV identification scheme are manifested as follows.

A. RF Signal Acquisition and Segmentation

The RF signals within the frequency band of UAV operation can be continuously acquired using a monitoring receiver as illustrated by Fig. 1 and the example of a captured UAV RF signal can be visualized in Fig. 2.

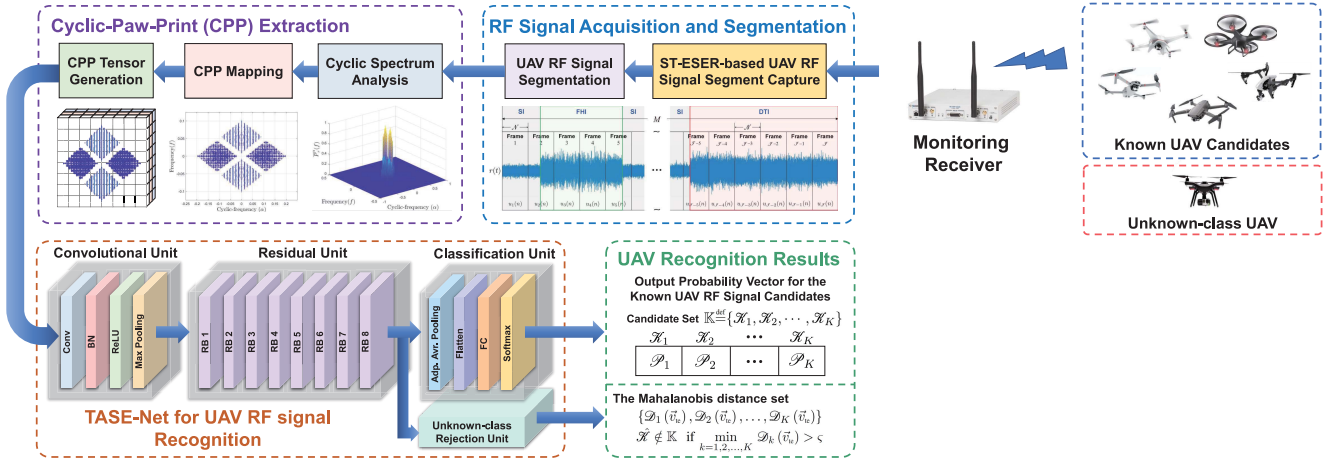


Fig. 1. Block diagram of our proposed novel robust open-set few-shot UAV identification technique.

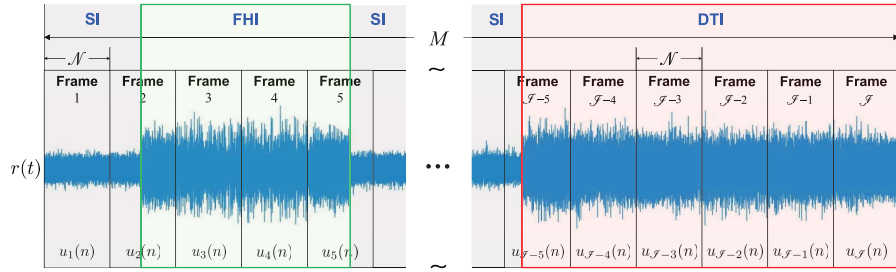


Fig. 2. Example of segmentation of a captured discrete-time UAV RF signal sequence denoted by $r(t)$.

According to Fig. 2, the acquired RF signal waveform can be split into “data-transmission intervals (DTIs),” “frequency-hopping intervals (FHIs),” and “silent intervals (SIs),” where DTI and FHI usually correspond to the digital data-transmission (DDT) signal and the flight control signal, respectively. Note that we only need to extract the features from the DTIs and FHIs for UAV identification. Thus, a sophisticated mechanism for detection and segmentation of a captured RF signal is designed in this work.

Let’s denote a captured M -sample discrete-time RF signal sequence by $r(t)$, $t = 0, 1, \dots, M - 1$, which can be continuously and seamlessly segmented using a sliding window, which consists of $\mathcal{N}$ consecutive samples. Now partition $r(t)$ into $\mathcal{F}$ frames (windows) as depicted by Fig. 2, where $\mathcal{F} = M/\mathcal{N}$ and the i^{th} frame of signal samples can be expressed by $\{u_i(t'), t' = 0, 1, \dots, \mathcal{N} - 1\}$ such that

$$u_i(t') \stackrel{\text{def}}{=} r(t' + (i - 1)\mathcal{N}), \quad t' = 0, 1, \dots, \mathcal{N} - 1 \quad (1)$$

where $i = 1, 2, \dots, \mathcal{F}$. The discrete Fourier transform (DFT) sequence of the i^{th} frame of signal samples $\{u_i(t'), t' = 0, 1, \dots, \mathcal{N} - 1\}$ for $i = 1, 2, \dots, \mathcal{F}$ can be obtained as

$$U_i(m) \stackrel{\text{def}}{=} \sum_{t'=0}^{\mathcal{N}-1} u_i(t') \exp\left(-\frac{j2\pi t' m}{\mathcal{N}}\right), \quad m = 0, 1, \dots, \mathcal{N} - 1. \quad (2)$$

Then, the magnitude spectrum sequence of the i^{th} frame can be computed as $|U_i(m)|$, $m = 0, 1, \dots, \mathcal{N} - 1$. Now, according to [33], we would like to form a “spectral probability

sequence” $P_i(m)$ of the i^{th} frame such that

$$P_i(m) \stackrel{\text{def}}{=} \frac{|U_i(m)|^2}{\sum_{v=0}^{\mathcal{N}-1} |U_i(v)|^2}, \quad m = 0, 1, \dots, \mathcal{N} - 1. \quad (3)$$

According to [33], the corresponding *spectral entropy* of the i^{th} frame based on (3) can be obtained as

$$H_i \stackrel{\text{def}}{=} -\sum_{m=0}^{\mathcal{N}-1} P_i(m) \ln[P_i(m)]. \quad (4)$$

We define the “ST-ESER” e_i of the i^{th} frame by

$$e_i \stackrel{\text{def}}{=} \frac{\sum_{m=0}^{\mathcal{N}-1} |U_i(m)|^2}{H_i}. \quad (5)$$

The collection of ST-ESERs of all $\mathcal{F}$ frames can form a set $\mathbb{E} \stackrel{\text{def}}{=} \{e_1, e_2, \dots, e_{\mathcal{F}}\}$. Note that $e_i \gg 1$ in the presence of a UAV signal; the larger the ST-ESER e_i , the more confident we are that a UAV signal does appear in the i^{th} frame. As shown by Fig. 3, large ST-ESER values can be observed in the presence of drone signals. Let’s denote the random variable corresponding to e_i by E_i , $i = 1, 2, \dots, \mathcal{F}$. Without loss of generality, we further assume that $E_1, E_2, \dots, E_{\mathcal{F}}$ are independently and identically distributed (i.i.d.) such that the corresponding probability density functions (PDFs) $\ell_{E_i}(e_i)$, $i = 1, 2, \dots, \mathcal{F}$ can be represented by

$$\ell_E(e) \stackrel{\text{def}}{=} \ell_{E_i}(e_i) \text{ for } i = 1, 2, \dots, \mathcal{F}. \quad (6)$$

Thus, the smallest, second smallest, third smallest, ..., and the largest ST-ESERs can be represented by the random variables:

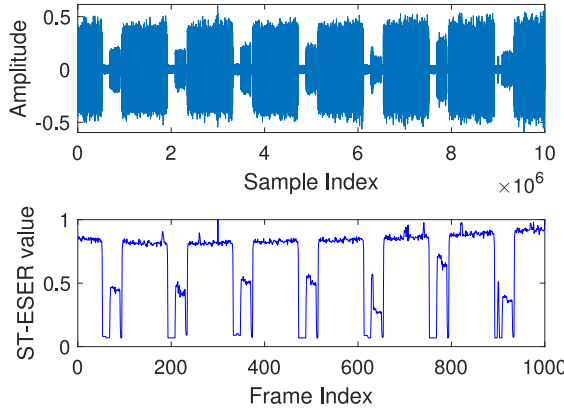


Fig. 3. Signal waveform and the corresponding frame ST-ESERs of the DJI MATRICE 600 Pro drone in [34].

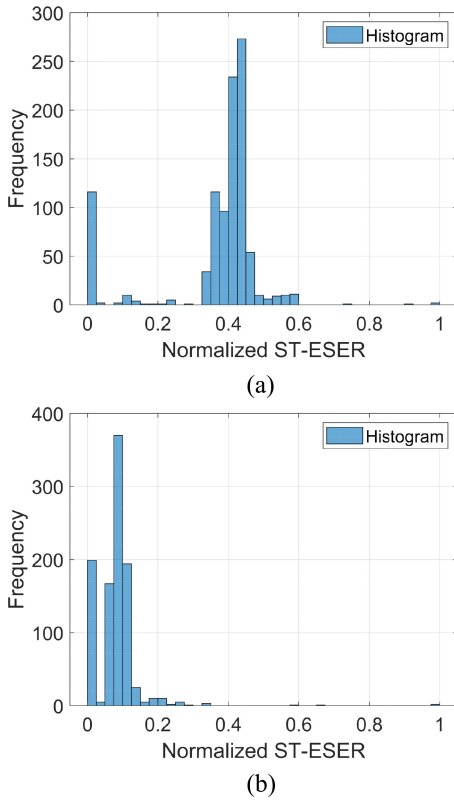


Fig. 4. Sample histograms $\hat{\mathcal{F}}_E(e)$ of the DTI signal sequences emitted by two typical DJI UAV models. (a) $\hat{\mathcal{F}}_E(e)$ for DJI MATRICE 600 Pro 2. (b) $\hat{\mathcal{F}}_E(e)$ for DJI Inspire 2.

$E_{(1)}, E_{(2)}, E_{(3)}, \dots, E_{(\mathcal{F})}$ such that $E_{(1)} \leq E_{(2)} \leq \dots \leq E_{(\mathcal{F})}$. Thus, the empirical cumulative distribution function (ECDF) $\mathcal{F}_E(e)$ of the PDF given by (6) can be estimated using the corresponding sample histogram of e_i 's. The sample histograms $\hat{\mathcal{F}}_E(e)$ of two types of UAV DJI signal sequences (emitted by DJI MATRICE 600 Pro 2 and DJI Inspire 2 models) are illustrated by Fig. 4.

According to the ECDF estimate $\hat{\mathcal{F}}_E(e)$ (using the sample histogram), the ECDF of $E_{(\iota)}$ (the ι^{th} order statistic (see [35]) or the ι^{th} smallest ST-ESER among $E_1, E_2, \dots, E_{\mathcal{F}}$) can be established by

$$\begin{aligned} P[E_{(\iota)} \leq e] &\stackrel{\text{def}}{=} \mathcal{F}_{E_{(\iota)}}(e) \\ &= \sum_{j=\iota}^{\mathcal{F}} \binom{\mathcal{F}}{j} [\mathcal{F}_E(e)]^j [1 - \mathcal{F}_E(e)]^{\mathcal{F}-j} \end{aligned} \quad (7)$$

where $\iota = 1, 2, \dots, \mathcal{F}$ and $\binom{\mathcal{F}}{j} \stackrel{\text{def}}{=} ([\mathcal{F}!]/([(\mathcal{F}-j)!j!])$. For a given confidence level, say ϱ (in percentage) and a given ST-ESER threshold $\bar{e}$, the appropriate ι can be determined by solving the following equation:

$$P[E_{(\iota)} > \bar{e}] = 1 - P[E_{(\iota)} \leq \bar{e}] = 1 - \mathcal{F}_{E_{(\iota)}}(\bar{e}) = \varrho. \quad (8)$$

Without loss of generality, the confidence level ϱ is set to be 0.95 and the threshold $\bar{e}$ is set to be the most probable ST-ESER value (the mode of ST-ESER).

As previously mentioned, a captured M -sample discrete-time RF signal sequence by $r(t)$, $t = 0, 1, \dots, M-1$ will be segmented into $\mathcal{F}$ frames of signal samples, i.e., $\{u_i(t'), t' = 0, 1, \dots, \mathcal{N}-1\}$ for $i = 1, 2, \dots, \mathcal{F}$. Then, we will retain (select) the ϕ frames (indexed by $i = \pi(1), \pi(2), \dots, \pi(\phi)$) of signal samples corresponding to the ϕ largest ST-ESERs, where

$$\phi \stackrel{\text{def}}{=} \mathcal{F} - \iota + 1 \quad (9)$$

such that the retained signal samples form a new signal sequence $x(n)$, $n = 0, 1, \dots, \phi\mathcal{N}-1$ as given by

$$x(n) \stackrel{\text{def}}{=} u_{\pi(\vartheta)}(n - \vartheta\mathcal{N}), \quad n = 0, 1, \dots, \phi\mathcal{N}-1 \quad (10)$$

where $\vartheta \stackrel{\text{def}}{=} \lfloor n/\mathcal{N} \rfloor$ and “ $\lfloor \cdot \rfloor$ ” specifies the integer rounding-down (floor) operation.

B. CPP Extraction

According to our previous studies in [30], [31], and [32], different periodicities of the second-order statistics of different UAV control and communication signals have been explored to undertake robust ACMC. Henceforth, the SCS analysis is also invoked here to characterize the sequence $x(n)$, $n = 0, 1, \dots, \phi\mathcal{N}-1$ given by (10) in the polyspectral domain. The second-order cyclic autocorrelation function of $x(n)$ can be derived by utilizing the fast Fourier transform (FFT) accumulation method (FAM) according to [36] and the time-smoothed cyclic periodogram $\mathcal{S}_{xT}^{\alpha}(n, f)_{\Delta t}$ for a given frequency f and a cycle frequency α is given by

$$\mathcal{S}_{xT}^{\alpha}(n, f)_{\Delta t} \stackrel{\text{def}}{=} \sum_{\lambda} \mathcal{X}_T(\lambda, f_1) \mathcal{X}_T^*(\lambda, f_2) g'(n - \lambda) \quad (11)$$

where $g'(n)$ denotes a unity-area weighting function of width $\Delta t \stackrel{\text{def}}{=} NT_s$ seconds, N denotes the number of samples per FFT window, T_s denotes the sampling period, f_1 and f_2 denote the center frequencies of the filters used in the FAM, i.e., $f_1 \stackrel{\text{def}}{=} f + \alpha/2$, $f_2 \stackrel{\text{def}}{=} f - \alpha/2$. According to [36], $\mathcal{X}_T(\lambda, f_1)$, $\mathcal{X}_T(\lambda, f_2)$ in (11) are the complex demodulates of $x(n)$. That is

$$\mathcal{X}_T(\lambda, f) \stackrel{\text{def}}{=} \sum_{\lambda=-N'/2}^{N'/2-1} \hat{w}(\lambda) x(n - \lambda) e^{-j2\pi f(n-\lambda)T_s} \quad (12)$$

where $\hat{w}(\lambda)$ denotes a data-tapering window of duration $T \stackrel{\text{def}}{=} N'T_s$ seconds and its bandwidth coincides with the frequency

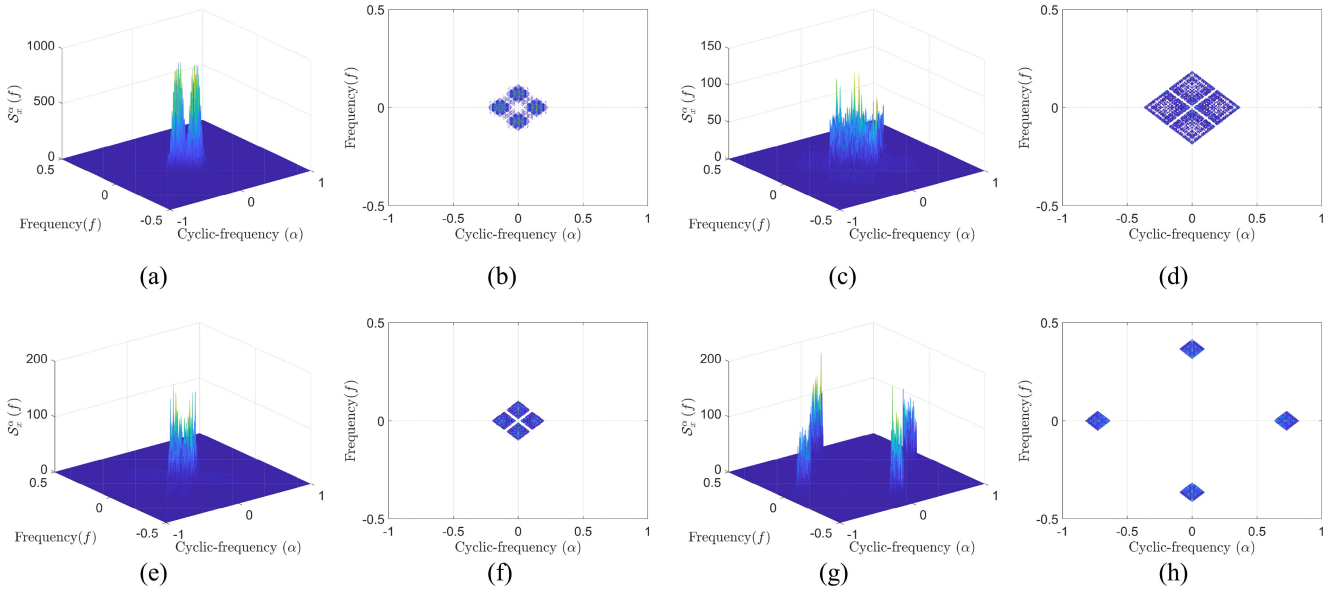


Fig. 5. Illustration of the NSCSs and the corresponding top views (CPPs) of certain typical UAV RF signals presented in [34]. (a) NSCS of DJI Phantom 400 Pro. (b) Top view of (a). (c) NSCS of DJI Mavic Pro. (d) Top view of (c). (e) NSCS of DJI Mini 2. (f) Top view of (e). (g) NSCS of DJI MATRICE 600 Pro. (h) Top view of (g).

resolution $\Delta f = f_s/N'$ of the SCS. Note that f_s denotes the sampling frequency. According to [32], the SCS can be estimated from the time-smoothed cyclic periodogram $\mathcal{S}_{x_T}^\alpha(n, f)_{\Delta t}$ without bias such that

$$\mathcal{S}_x^\alpha(f) \stackrel{\text{def}}{=} \lim_{\Delta f \rightarrow 0} \lim_{\Delta t \rightarrow 0} \mathcal{S}_{x_T}^\alpha(n, f)_{\Delta t}. \quad (13)$$

The SCS $\mathcal{S}_x^\alpha(f)$, which contains $N' + 1$ spectral frequencies $f = f_q$ for $q = -N'/2, -N'/2 + 1, \dots, N'/2$ and $2N + 1$ cycle frequencies $\alpha = \alpha_p$ for $p = -N, -N + 1, \dots, N$, is a 3-D spectrum with nonnegative amplitudes. According to [32], the NSCS can be computed as

$$\tilde{\mathcal{S}}_x^\alpha \stackrel{\text{def}}{=} \frac{\mathcal{S}_x^\alpha(f)}{\max_{\alpha, f} \mathcal{S}_x^\alpha(f)}. \quad (14)$$

The NSCSs of the typical UAV RF signals emitted by the UAVs specified in [34] and the corresponding top views are illustrated by Fig. 5.

According to Fig. 5, the top view of the NSCS of a UAV RF signal would often be concentrated in four symmetric regions of the $\alpha - f$ plane while the corresponding scatter plot resembles an animal's paw print. Thus, this scatter plot was named after "CPP" in [30], [31], and [32]. It is apparent from Fig. 5 that the CPPs resulting from different UAV RF signals are quite distinguishable, which can thus serve as robust features for UAV identification. The top view of an NSCS can be represented by a $P \times Q$ matrix $\tilde{\mathcal{S}}_x \stackrel{\text{def}}{=} [\tilde{\mathcal{S}}_x(p, q)]_{0 \leq p \leq P-1, 0 \leq q \leq Q-1}$, where $P \stackrel{\text{def}}{=} 2N + 1$ and $Q \stackrel{\text{def}}{=} N' + 1$. The entry $\tilde{\mathcal{S}}_x(p, q)$ in $\tilde{\mathcal{S}}_x$ is the nonnegative and normalized amplitude of $\mathcal{S}_x^\alpha(f)$ at the cyclic frequency α_p and the spectral frequency f_q for $p = p + N$ and $q = q + N'/2$. According to [32], the top view of an NSCS $\tilde{\mathcal{S}}_x$ can be treated as a grayscale colormap (black and white image)

and further converted to a "quantized" image matrix $\tilde{\mathcal{S}}_x' \stackrel{\text{def}}{=} [\mathcal{S}_{p,q}]_{0 \leq p \leq P-1, 0 \leq q \leq Q-1}$ by a b -bit quantizer such that

$$\mathcal{S}_{p,q} \stackrel{\text{def}}{=} \left\lceil 2^{b-1} \times \left| \tilde{\mathcal{S}}_x(p, q) \right| \right\rceil, \quad p = 1, 2, \dots, P, \quad q = 1, 2, \dots, Q \quad (15)$$

where b is a positive integer. Without loss of generality, b can be set to be 16 in this work.

The grayscale image $\tilde{\mathcal{S}}_x'$ is reshaped into a $\Lambda \times \Lambda$ square matrix $\tilde{\mathcal{T}} \stackrel{\text{def}}{=} [\mathcal{S}_{p,q}^\dagger]_{1 \leq p \leq \Lambda, 1 \leq q \leq \Lambda}$ by decimating each column vector (reducing $N' + 1$ entries to Λ entries for each column of $\tilde{\mathcal{S}}_x'$) and interpolating each row vector (extending $2N + 1$ entries to Λ entries for each row of $\tilde{\mathcal{S}}_x'$) subject to $P < \Lambda < Q$. Then, the square grayscale colormap matrix $\tilde{\mathcal{T}}$ can be further converted to a pseudo-color square image, namely, the "parula colormap" using the corresponding function in MATLAB (see [37]), which can be expressed by

$$\mathcal{S}_{p,q}^\dagger \xrightarrow{\text{parula}} [\mathcal{T}_{p,q}^R, \mathcal{T}_{p,q}^G, \mathcal{T}_{p,q}^B]^T \quad p = 1, 2, \dots, \Lambda \text{ and } q = 1, 2, \dots, \Lambda. \quad (16)$$

According to (16), the CPP, which is the square pseudo-color image, can be represented by the following third-order tensor:

$$\mathcal{T}_{\text{RGB}} \stackrel{\text{def}}{=} [\tilde{\mathcal{T}}_R, \tilde{\mathcal{T}}_G, \tilde{\mathcal{T}}_B] \quad (17)$$

where $\tilde{\mathcal{T}}_R$, $\tilde{\mathcal{T}}_G$, and $\tilde{\mathcal{T}}_B$ denote three square matrices for the individual redness (R), greenness (G), and blueness (B) channels such that

$$\tilde{\mathcal{T}}_R \stackrel{\text{def}}{=} [\mathcal{T}_{p,q}^R]_{1 \leq p \leq \Lambda, 1 \leq q \leq \Lambda} \quad (18)$$

$$\tilde{\mathcal{T}}_G \stackrel{\text{def}}{=} [\mathcal{T}_{p,q}^G]_{1 \leq p \leq \Lambda, 1 \leq q \leq \Lambda} \quad (19)$$

$$\tilde{\mathcal{T}}_B \stackrel{\text{def}}{=} [\mathcal{T}_{p,q}^B]_{1 \leq p \leq \Lambda, 1 \leq q \leq \Lambda}. \quad (20)$$

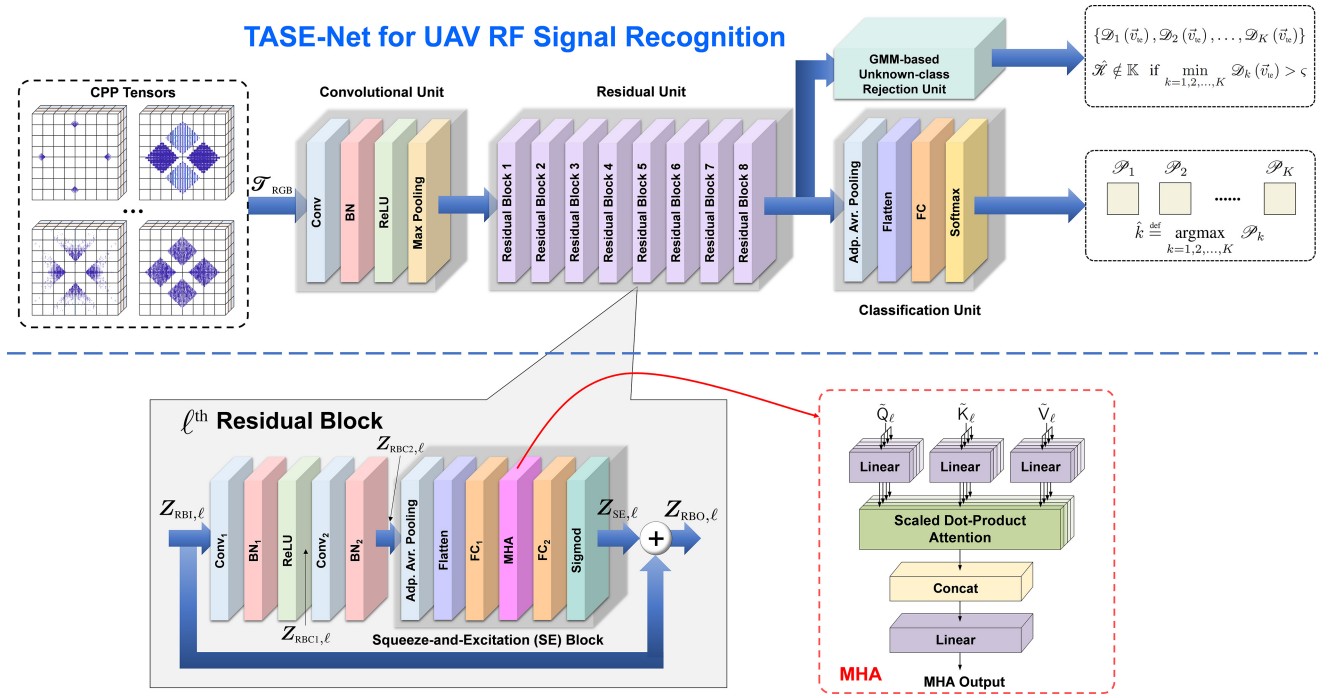


Fig. 6. Overall architecture of our proposed TASE-Net for open-set few-shot UAV recognition where “Conv,” “BN,” “Adp. Avr. Pooling,” “ReLU,” “FC,” “MHA,” and “Concat” denote “convolutional layer,” “batch norm layer,” “adaptive average pooling layer,” “rectified linear unit,” “fully connected layer,” “multihead attention layer,” and “concatenation,” respectively.

C. Transformer-Attention Squeeze-Excitation Network

Once the CPP tensor $\mathcal{T}_{RGB}$ of a received UAV RF signal is built according to (17)–(20), our proposed novel TASE-Net can be trained to identify the UAV RF signals. Our proposed TASE-Net is designed by incorporating the transformer-enhanced SE model into the ResNet, and a new unknown-class detection scheme using the GMM is also included in the TASE-Net for open-set classification. Our proposed TASE-Net can not only lead to a promising classification performance for the open-set few-shot scenario but also require a low computational complexity. The overall architecture of our proposed TASE-Net is illustrated by Fig. 6, which consists of a convolutional unit, eight residual blocks, a classification unit, and an unknown-class rejection unit. The parameter settings of the complete architecture of our proposed TASE-Net can be found in the supplementary document. Now let’s introduce the details of TASE-Net as follows.

1) *Convolution Unit*: In the convolutional unit as shown in Fig. 6, the convolutional layer involving sixty-four filters (kernels) of size 7×7 with the stride of 2 and the padding of 3 is adopted to extract essential features from a CPP image (tensor), which is followed by a batch normalization (BN) layer, a rectified linear unit (ReLU) activation layer, and a max-pooling layer. The third-order CPP tensor of a captured UAV RF signal, namely, $\mathcal{T}_{RGB}$, is directly fed to the convolutional unit. The corresponding output is given by

$$\mathbf{Z}_c \stackrel{\text{def}}{=} \mathbf{W}_c \otimes \mathcal{T}_{RGB} + \mathbf{B}_c \quad (21)$$

where $\mathbf{W}_c$ and $\mathbf{B}_c$ denote the kernel filter and the bias for the convolutional layer (see [38]), respectively. Then, $\mathbf{Z}_c$ is normalized with respect to its mean μ_c and variance σ_c^2 at

the following BN layer and a ReLU activation function is applied to the normalized result to accelerate the convergence and improve the classification accuracy. Finally, the spatial dimensions of the output of ReLU are downsampled by a max-pooling layer, and the output of the convolutional unit denoted by $\hat{\mathbf{Z}}_c$ is fed to the following residual unit. Please refer to the supplementary document for the detailed structural description of the convolutional unit depicted in Fig. 6.

2) *Residual Unit*: The residual unit is composed by eight residual blocks (see [39]) which share the identical structure as shown in Fig. 6. Each residual block consists of two convolutional layers (labeled by “Conv₁” and “Conv₂” in Fig. 6), two batch norm layers (labeled by “BN₁” and “BN₂” in Fig. 6), a ReLU layer, and an SE block [40], where the transformer-enhanced SE is incorporated into each residual block to facilitate the feature-map recognition. The input and output of the ℓ^{th} residual block, $\ell = 1, 2, \dots, 8$, denoted by $\mathbf{Z}_{RBI,\ell}$ and $\mathbf{Z}_{RBO,\ell}$, respectively, is processed by the first convolutional layer (Conv₁), the first batch norm layer (BN₁), and a ReLU activation function, where $\mathbf{Z}_{RBI,1} = \hat{\mathbf{Z}}_c$ for the first residual block ($\ell = 1$), and $\mathbf{Z}_{RBI,\ell} = \mathbf{Z}_{RBO,\ell-1}$ for $\ell = 2, 3, \dots, 8$. Thus, according to Fig. 6, the normalized and activated output in the ℓ^{th} residual block (see [39]) can be expressed by

$$\mathbf{Z}_{RBC1,\ell} \stackrel{\text{def}}{=} \text{ReLU} \left\{ \text{BN}_1 \left[\mathbf{Z}_{RBI,\ell} \otimes \mathbf{W}_{RBC1,\ell} + \mathbf{B}_{RBC1,\ell} \right] \right\} \quad \ell = 1, 2, \dots, 8 \quad (22)$$

where $\mathbf{W}_{RBC1,\ell}$ and $\mathbf{B}_{RBC1,\ell}$ denote the kernel filter and the bias of “Conv₁” in the ℓ^{th} residual block, respectively. Then, $\mathbf{Z}_{RBC1,\ell}$ involved in (22) is further processed by the second

convolutional layer (“Conv₂”) and the second batch norm layer (“BN₂”) to produced the corresponding output

$$\mathbf{Z}_{\text{RBC2},\ell} = \text{BN}_2 \left[\mathbf{Z}_{\text{RBC1},\ell} \otimes \mathbf{W}_{\text{RBC2},\ell} + \mathbf{B}_{\text{RBC2},\ell} \right] \quad (23)$$

where $\mathbf{W}_{\text{RBC2},\ell}$ and $\mathbf{B}_{\text{RBC2},\ell}$ denote the kernel filter and the bias of “Conv₂” in the ℓ^{th} residual block, respectively.

Next, $\mathbf{Z}_{\text{RBC2},\ell}$ is sent to our proposed transformer-enhanced SE block, which consists of an adaptive average pooling layer, a flatten layer, two fully connected layers (FC₁ and FC₂), a multihead attention layer (MHA), and a sigmoid function as illustrated by Fig. 6. Please refer to the supplementary document for the detailed structural description of the residual unit depicted in Fig. 6. According to Fig. 6, the ℓ^{th} residual block takes $\mathbf{Z}_{\text{RBC2},\ell}$ as the input of the SE block to produce the output $\mathbf{Z}_{\text{SE},\ell}$, $\ell = 1, 2, \dots, 8$.

As illustrated by Fig. 6, the ultimate output $\mathbf{Z}_{\text{RBO},\ell}$ of the ℓ^{th} residual block is given by

$$\mathbf{Z}_{\text{RBO},\ell} \stackrel{\text{def}}{=} \mathbf{Z}_{\text{RBI},\ell} + \mathbf{Z}_{\text{SE},\ell}, \quad \ell = 1, 2, \dots, 8. \quad (24)$$

According to Fig. 6, the output of the last (eighth) residual block, namely, $\mathbf{Z}_{\text{RBO},8}$ is simultaneously sent to the classification unit and the (GMM-based) unknown-class rejection unit for recognizing or rejecting the received UAV RF signal.

3) *Classification Unit*: As illustrated by Fig. 6, the output of the last (eighth) residual block, $\mathbf{Z}_{\text{RBO},8}$ is fed to the classification unit, which is composed by an adaptive average pooling layer, a flatten layer, and a fully connected layer in series, and it will produce a feature vector $\tilde{\rho} \stackrel{\text{def}}{=} [\rho_1, \rho_2, \dots, \rho_K]^T$, where K corresponds to the total number of known UAV RF candidates for training and $\sum_{k=1}^K \rho_k = 1$. Then, the probability of the received signal matching the k^{th} type of known UAV RF signal ($k = 1, 2, \dots, K$) in the candidate set $\mathbb{K} \stackrel{\text{def}}{=} \{\mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_K\}$ ($|\mathbb{K}| = K$) can be estimated by a “softmax” activation function such that

$$\mathcal{P}_k \stackrel{\text{def}}{=} \text{softmax}(\rho_k), \quad k = 1, 2, \dots, K. \quad (25)$$

Finally, the type of the received RF signal, say $\hat{\mathcal{K}}$, can be identified as

$$\hat{\mathcal{K}} \stackrel{\text{def}}{=} \mathcal{K}_{\hat{k}} \quad (26)$$

where

$$\hat{k} \stackrel{\text{def}}{=} \underset{k=1,2,\dots,K}{\text{argmax}} \mathcal{P}_k. \quad (27)$$

4) *Unknown-Class Rejection Unit*: Furthermore, according to Fig. 6, the output of the flatten layer in the classification unit, namely, $\tilde{\mathbf{v}} \stackrel{\text{def}}{=} [v_1, v_2, \dots, v_D]^T$, is also sent to the (GMM-based) unknown-class rejection unit to determine if the type of the received UAV RF signal belongs to the training candidate set $\mathbb{K}$. Our proposed unknown-class rejection unit will measure the discrepancy between the feature tensor extracted from the received UAV RF signal and that extracted from each of the known candidates in $\mathbb{K}$ in terms of *Mahalanobis distance*. Such a Mahalanobis distance will be further compared with an appropriate threshold to determine if a candidate signal type shall be rejected. According to [41], $\tilde{\mathbf{v}}$ can be regarded as a random vector with the K -mode Gaussian-mixture PDF where

K specifies the total number of known UAV RF signals for training. Such a PDF $f_{\tilde{\mathbf{v}}}(\tilde{\mathbf{v}})$ is characterizes as

$$f_{\tilde{\mathbf{v}}}(\tilde{\mathbf{v}}) \stackrel{\text{def}}{=} \sum_{k=1}^K \xi_k \Psi(\tilde{\mathbf{v}} | \tilde{\boldsymbol{\mu}}_k, \tilde{\boldsymbol{\Gamma}}_k) \quad (28)$$

where ξ_k denotes the weighting coefficient of the k^{th} Gaussian mode, $0 \leq \xi_k \leq 1$, and $\sum_{k=1}^K \xi_k = 1$. Note that

$$\Psi(\tilde{\mathbf{v}} | \tilde{\boldsymbol{\mu}}_k, \tilde{\boldsymbol{\Gamma}}_k) \stackrel{\text{def}}{=} \frac{1}{(2\pi)^{D/2} |\tilde{\boldsymbol{\Gamma}}_k|^{1/2}} \times \exp \left\{ -\frac{1}{2} (\tilde{\mathbf{v}} - \tilde{\boldsymbol{\mu}}_k)^T \tilde{\boldsymbol{\Gamma}}_k^{-1} (\tilde{\mathbf{v}} - \tilde{\boldsymbol{\mu}}_k) \right\} \quad (29)$$

where $\tilde{\boldsymbol{\mu}}_k \in \mathbb{R}^{D \times 1}$ and $\tilde{\boldsymbol{\Gamma}}_k \in \mathbb{R}^{D \times D}$ denote the corresponding mean vector and covariance matrix, respectively.

Once the set of parameters $\Theta \stackrel{\text{def}}{=} \bigcup_{k=1}^K \theta_k = \{\xi_k, \tilde{\boldsymbol{\mu}}_k, \tilde{\boldsymbol{\Gamma}}_k\}_{k=1}^K$ are estimated from the training data, the K -mode Gaussian mixture PDF can be characterized, where $\theta_k \stackrel{\text{def}}{=} \{\xi_k, \tilde{\boldsymbol{\mu}}_k, \tilde{\boldsymbol{\Gamma}}_k\}$ consists of the parameters associated with the k^{th} Gaussian mode, $k = 1, 2, \dots, K$. Assume that for each training candidate $\mathcal{K}_k$, $k = 1, 2, \dots, K$, we can collect ζ output samples, namely, $\tilde{\mathbf{v}}_{\text{tr},k,1}, \tilde{\mathbf{v}}_{\text{tr},k,2}, \dots, \tilde{\mathbf{v}}_{\text{tr},k,\zeta}$ of the flatten layer in the classification unit. We propose to apply the well-known expectation-maximization algorithm in [42], [43], and [44] to estimate Θ as follows.

Initialization: Estimate ξ_k , $\tilde{\boldsymbol{\mu}}_k$, and $\tilde{\boldsymbol{\Gamma}}_k$ for $k = 1, 2, \dots, K$ such that

$$\hat{\xi}_{\text{pre},k} \stackrel{\text{def}}{=} 1/K \quad (30)$$

$$\hat{\tilde{\boldsymbol{\mu}}}_{\text{pre},k} \stackrel{\text{def}}{=} \frac{1}{\zeta} \sum_{\ell=1}^{\zeta} \tilde{\mathbf{v}}_{\text{tr},k,\ell} \quad (31)$$

$$\hat{\tilde{\boldsymbol{\Gamma}}}_{\text{pre},k} \stackrel{\text{def}}{=} \frac{1}{\zeta} \sum_{\ell=1}^{\zeta} \left[\tilde{\mathbf{v}}_{\text{tr},k,\ell} - \hat{\tilde{\boldsymbol{\mu}}}_{\text{pre},k} \right] \left[\tilde{\mathbf{v}}_{\text{tr},k,\ell} - \hat{\tilde{\boldsymbol{\mu}}}_{\text{pre},k} \right]^T \quad (32)$$

for $k = 1, 2, \dots, K$.

Step 1 (Expectation Step): Re-estimate ξ_k for the current (new) iteration as $\hat{\xi}_{\text{cur},k}$ such that

$$\gamma_{k,\varepsilon,\ell} \stackrel{\text{def}}{=} \frac{1}{\sqrt{|\tilde{\boldsymbol{\Gamma}}_{\text{pre},k}|}} \times \exp \left\{ -\frac{1}{2} \left(\tilde{\mathbf{v}}_{\text{tr},\varepsilon,\ell} - \hat{\tilde{\boldsymbol{\mu}}}_{\text{pre},k} \right)^T \times \tilde{\boldsymbol{\Gamma}}_{\text{pre},k}^{-1} \left(\tilde{\mathbf{v}}_{\text{tr},\varepsilon,\ell} - \hat{\tilde{\boldsymbol{\mu}}}_{\text{pre},k} \right) \right\} \quad (33)$$

$$\hat{\xi}_{\text{cur},k} \stackrel{\text{def}}{=} \frac{\sum_{\varepsilon=1}^K \sum_{\ell=1}^{\zeta} \gamma_{k,\varepsilon,\ell}}{\sum_{k'=1}^K \sum_{\varepsilon=1}^K \sum_{\ell=1}^{\zeta} \gamma_{k',\varepsilon,\ell}} \quad \text{for } k = 1, 2, \dots, K. \quad (34)$$

Step 2 (Maximization Step): Re-estimate $\tilde{\boldsymbol{\mu}}_k$ and $\tilde{\boldsymbol{\Gamma}}_k$ as $\hat{\tilde{\boldsymbol{\mu}}}_{\text{cur},k}$ and $\hat{\tilde{\boldsymbol{\Gamma}}}_{\text{cur},k}$, respectively, $k = 1, 2, \dots, K$ for the current (new) iteration such that

$$\hat{\tilde{\boldsymbol{\mu}}}_{\text{cur},k} \stackrel{\text{def}}{=} \sum_{\varepsilon=1}^K \sum_{\ell=1}^{\zeta} \gamma_{k,\varepsilon,\ell} \times \tilde{\mathbf{v}}_{\text{tr},\varepsilon,\ell} \quad (35)$$

$$\hat{\tilde{\boldsymbol{\Gamma}}}_{\text{cur},k} \stackrel{\text{def}}{=} \sum_{\varepsilon=1}^K \sum_{\ell=1}^{\zeta} \gamma_{k,\varepsilon,\ell} \times \left[\tilde{\mathbf{v}}_{\text{tr},\varepsilon,\ell} - \hat{\tilde{\boldsymbol{\mu}}}_{\text{cur},k} \right] \left[\tilde{\mathbf{v}}_{\text{tr},\varepsilon,\ell} - \hat{\tilde{\boldsymbol{\mu}}}_{\text{cur},k} \right]^T \quad (36)$$

for $k = 1, 2, \dots, K$.

Step 3: Update

$$\hat{\xi}_{\text{pre},k} \leftarrow \hat{\xi}_{\text{cur},k}, \hat{\mu}_{\text{pre},k} \leftarrow \hat{\mu}_{\text{cur},k}, \hat{\Gamma}_{\text{pre},k} \leftarrow \hat{\Gamma}_{\text{cur},k} \quad (37)$$

for $k = 1, 2, \dots, K$. Go back to Step 1 until the convergence of all parameter estimates and the set of final (converged) parameter estimates is expressed by $\{\hat{\xi}_k, \hat{\mu}_k, \hat{\Gamma}_k\}_{k=1}^K$.

During test, the output of the flatten layer in the classification unit resulting from a received UAV RF signal can be expressed by $\tilde{v}_{\text{te}}$. The Mahalanobis distance (see [45]) between $\tilde{v}_{\text{te}}$ and the mean feature-vector of each UAV RF signal candidate in $\mathbb{K}$ can be calculated as

$$\mathcal{D}_k(\tilde{v}_{\text{te}}) = \sqrt{(\tilde{v}_{\text{te}} - \hat{\mu}_k)^T \hat{\Gamma}_k^{-1} (\tilde{v}_{\text{te}} - \hat{\mu}_k)}, \quad k = 1, 2, \dots, K. \quad (38)$$

Assume that the type of UAV RF signal to produce the test feature-vector $\tilde{v}_{\text{te}}$ is $\mathcal{K}$. One can choose a proper threshold, say ς , such that the rejection rule is given by

$$\mathcal{K} \notin \mathbb{K} \text{ if } \min_{k=1,2,\dots,K} \mathcal{D}_k(\tilde{v}_{\text{te}}) > \varsigma \quad (39)$$

where “ $\mathcal{K} \notin \mathbb{K}$ ” means that the feature-vector $\tilde{v}_{\text{te}}$ is produced by none of the candidate in $\mathbb{K}$. In other words, it is an unknown UAV RF signal.

III. SIMULATION

The performance of our proposed new TASE-Net is evaluated by Monte Carlo simulations under various conditions using the large-scale drone RF signal dataset: “DroneRFa” in [34]. We also compare three other existing open-set few-shot UAV RF identification methods in [18] and [22]. The DroneRFa dataset consists of twenty-four types of RF communication signals between drones and their flight controllers as well as a type of background signal in the absence of drones, which includes small and medium-sized drones. Here, we randomly select the RF signals produced by eight different UAVs, namely, DJI Phantom 4 Pro, DJI MATRICE 200, DJI Air 2S, DJI Inspire 2, DJI Mavic Pro, DJI Mini 2, DJI Phantom 4 Pro RTK, and DJI MATRICE 600 Pro. All UAV RF signals acquired from the DroneRFa dataset are regarded as the “noise-free” (clean) signals and the additive white Gaussian noise (AWGN) is artificially added to adjust the corresponding SNRs ranging from -15dB to 15dB . For each type of UAV, the emitted signal waveform is acquired at the sampling rate of 100 MSps according to [34]. The sample size is $M = 10^8$ and each UAV RF signal is further split into $\mathcal{T} = 10^4$ frames with equal frame length. Our proposed new ST-ESER-based signal segmentation mechanism is employed to extract at least 10^5 samples (to form the new signal sequence $x(n)$ as stated by (10)). Once $x(n)$ is obtained, the corresponding CPP tensors can be constructed. The FFT window size for generating the SCS in FAM and the sample size are $N' = 256$ and $N = 16384$, respectively. Then, the resulting top view image δ'_x turns out to be a 257×32769 matrix, which is further converted to a $300 \times 300 \times 3$ tensor $\mathcal{T}_{\text{RGB}}$. For each type of UAV, a hundred CPP tensors are extracted from the corresponding UAV RF signal subject to a given SNR, which are divided into

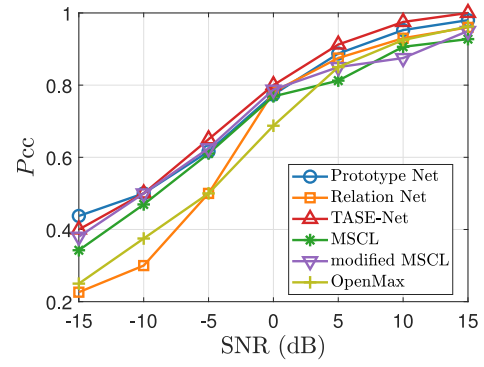


Fig. 7. P_{cc} 's (close-set classification) versus SNR for eight types of UAV RF signals in $\mathbb{K}$ resulting from our proposed TASE-Net and five other existing deep learning models in [18], [22], [46], and [47] using CPP features in the one-shot training scenario. The channel noise is assumed to be AWGN.

the training, validating, and testing datasets with respect to the ratio of 5:3:2. In our proposed TASE-Net, stochastic gradient descent (SGD) optimizer is invoked during training while the learning rate for updating each parameter is fixed to be 0.001. The cross-entropy loss-function is employed to calculate the training loss while the mini-batch size is set to be 5. The total number of training epochs is 50.

A. Close-Set Few-Shot UAV Classification

Let's investigate the close-set few-shot UAV classification first. All eight UAV RF signals are drawn from the DroneRFa dataset. Thus, we have $\mathbb{K} \stackrel{\text{def}}{=} \{\mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_8\}$. Our proposed novel UAV recognition approach is employed here to identify the received UAV RF signal from the known close-set $\mathbb{K}$ in the few-shot scenario. The corresponding classification performance is evaluated by the *probability of correct classification* (P_{cc}) over the entire UAV RF signal candidate set $\mathbb{K}$ and compared to the performances of other existing few-shot-learning models, namely, the MSCL model in [22], the modified MSCL model in [22], the OpenMax model in [18], the prototype network in [46], and the relation network in [47] using our proposed CPP features subject to various SNRs. For each SNR, CPP tensors are randomly acquired from the training dataset to train our proposed TASE-Net and other DL models in [18] and [22] while the number of shots per UAV type is set to be 1 and 5. The P_{cc} 's resulting from one and five shots of training are delineated by Figs. 7 and 8, respectively, in the presence of AWGN. It is conspicuous that more shots of training would lead to larger P_{cc} 's. According to Figs. 7 and 8, no matter in the one-shot or the five-shot scenario, our proposed TASE-Net always significantly outperforms other existing DL models for all SNR conditions.

B. Open-Set Few-Shot UAV Classification

Now let's investigate the open-set few-shot UAV classification. In this scenario, some of the aforementioned eight types of UAV RF signals are randomly picked to be the “unknown” UAV RF signals free from training. The metric “*openness*”¹

¹A different definition of openness can be found in [48] but our definition given by (40) is more informative.

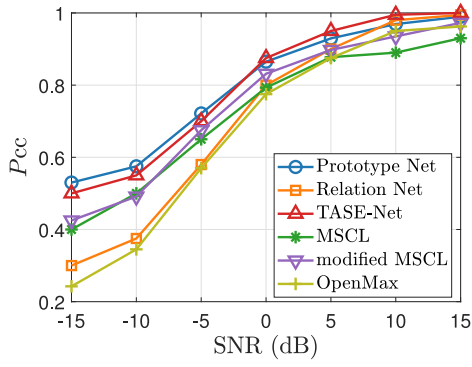


Fig. 8. P_{cc} 's (close-set classification) versus SNR for eight types of UAV RF signals in $\mathbb{K}$ resulting from our proposed TASE-Net and five other existing deep learning models in [18], [22], [46], and [47] using CPP features in the five-shot training scenario. The channel noise is assumed to be AWGN.

TABLE I
FIVE OPENNESS LEVELS FOR OUR SIMULATIONS

ψ (%)	K_{TR}	$K_{TE} - K_{TR}$
12.5	7	1
25.0	6	2
37.5	5	3
50.0	4	4
62.5	3	5

is defined here to measure the proportion of the test UAV RF signals to be unknown such that

$$\psi \stackrel{\text{def}}{=} 1 - \frac{K_{TR}}{K_{TE}} \quad (40)$$

where K_{TR} denotes the number of the types of known UAV RF signals in the training dataset and K_{TE} represents the total number of the types of UAV RF signals involved in the test. Consequently, $\psi = 0$ corresponds to the close-set scenario while $\psi > 0$ indicates the existence of at least an unknown UAV type during test. In this work, five openness levels are considered as listed in Table I where $K_{TE} - K_{TR}$ denotes the number of the types of unknown UAV RF signals.

Let's evaluate the unknown-class rejection performances resulting from all of the aforementioned schemes. In the absence of noise, we use the receiver-operating characteristic (ROC) curves² resulting from our proposed TASE-Net and three other existing DL models in [18] and [22] using CPP features for comparison. The area under curve (AUC) can be calculated from the corresponding ROC curves while the larger the AUC, the better the corresponding unknown-class rejection performance according to [49]. The AUCs with respect to ψ resulting from the five-shot training scenario are depicted by Fig. 9.

According to Fig. 9, our proposed TASE-Net always outperforms three other existing DL models in [18] and [22] for unknown-class rejection.

On the other hand, for those UAV RF signals drawn from the known types of UAVs, we also evaluate the P_{cc} 's and AUCs

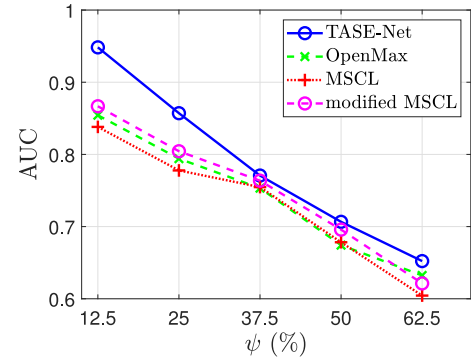


Fig. 9. AUCs for unknown-class rejection with respect to ψ resulting from our proposed TASE-Net and three other existing DL models in [18] and [22] using noise-free CPP features in the five-shot training scenario.

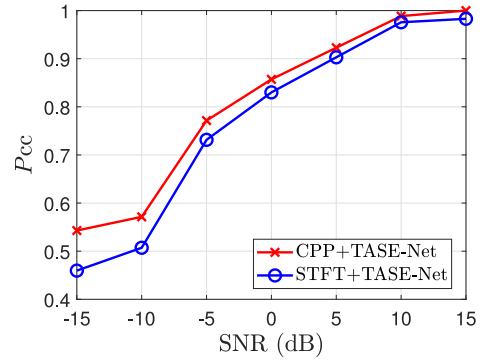


Fig. 10. P_{cc} 's (open-set classification) versus SNR resulting from our proposed TASE-Net using CPP and STFT features in the five-shot training scenario. The channel noise is assumed to be AWGN and $\psi = 12.5\%$.

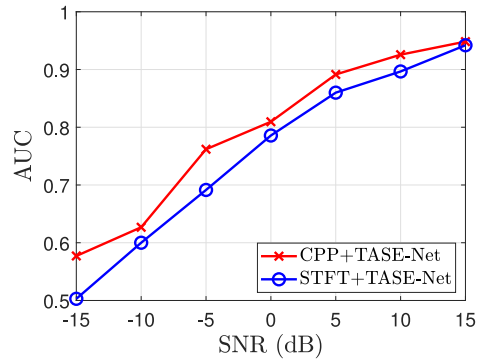


Fig. 11. AUCs for unknown-class rejection with respect to the SNR resulting from our proposed TASE-Net using CPP and STFT features subject to $\psi = 12.5\%$ in the five-shot training scenario. The channel noise is assumed to be AWGN and $\psi = 12.5\%$.

resulting from our proposed TASE-Net using CPP and short-time Fourier transform (STFT) features³ subject to various SNRs in the five-shot training scenario. Consider $\psi = 12.5\%$. The P_{cc} 's and AUCs resulting from our proposed TASE-Net using CPP and STFT features are delineated by Figs. 10 and 11.

According to Figs. 10 and 11, our proposed CPP features always achieve larger P_{cc} 's and AUCs than the STFT features.

²Please refer to [49] for the definition of ROC curve.

³The STFT features were proposed for UAV recognition in [34].

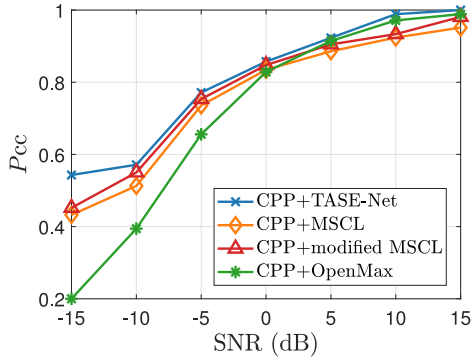


Fig. 12. P_{cc} 's (open-set classification) versus SNR resulting from our proposed TASE-Net and three other existing DL models in [18] and [22] using CPP feature in the five-shot training scenario. The channel noise is assumed to be AWGN and $\psi = 12.5\%$.

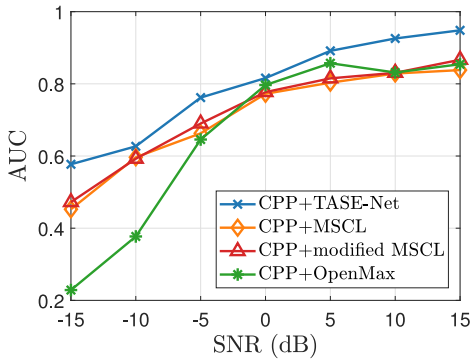


Fig. 13. AUCs for unknown-class rejection with respect to the SNR resulting from our proposed TASE-Net and three other existing DL models in [18] and [22] using CPP feature subject to $\psi = 12.5\%$ in the five-shot training scenario. The channel noise is assumed to be AWGN and $\psi = 12.5\%$.

Meanwhile, we also evaluate the P_{cc} 's and AUCs resulting from our proposed TASE-Net and three other existing DL models in [18] and [22] using CPP features subject to various SNRs and $\psi = 12.5\%$ in the five-shot training scenario are demonstrated by Figs. 12 and 13. According to Figs. 12 and 13, the P_{cc} 's and AUCs resulting from our proposed TASE-Net using CPP features are the largest among all schemes in comparison for all SNR conditions.

C. Numerical Parameter Determination

1) *Confidence Level ρ* : Here, we investigate the impact of the confidence level ρ in the ST-ESER-based frame selection process on the overall performance of our proposed novel TASE-Net. We consider several confidence levels: $\rho = 0.90$, 0.95 , and 0.99 while the ST-ESER threshold $\bar{e}$ is fixed to the mode of ST-ESERs. All other simulation settings remain unchanged as before. The performance of our proposed TASE-Net is evaluated subject to the aforementioned three confidence levels under different SNRs. The P_{cc} 's and AUCs with respect to $\psi = 12.5\%$ resulting from the five-shot training scenario are depicted by Figs. 14 and 15, respectively. According to Figs. 14 and 15, the P_{cc} and AUC curves resulting from $\rho = 0.95$ always lie above the P_{cc} and AUC curves resulting from $\rho = 0.90$ and $\rho = 0.99$. Consequently, the confidence level

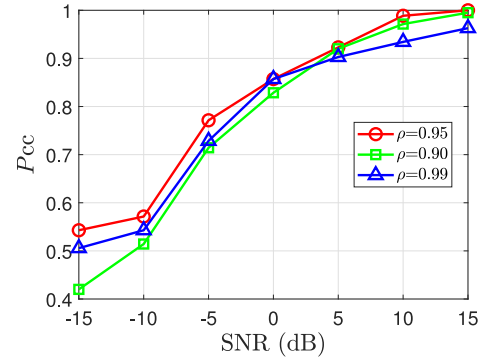


Fig. 14. P_{cc} 's (open-set classification) versus SNR resulting from our proposed TASE-Net subject to different confidence levels in the five-shot training scenario. The channel noise is assumed to be AWGN and $\psi = 12.5\%$.

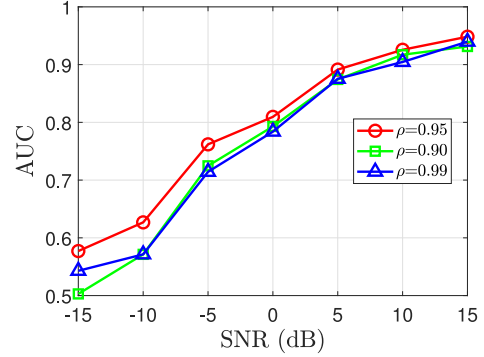


Fig. 15. AUCs for unknown-class rejection versus SNR resulting from our proposed TASE-Net subject to different confidence levels in the five-shot training scenario. The channel noise is assumed to be AWGN and $\psi = 12.5\%$.

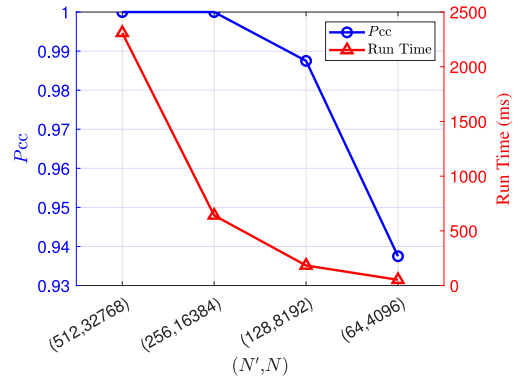


Fig. 16. P_{cc} 's and the run-times (in terms of milliseconds) with respect to various pairs of FFT window size N' and sample size N resulting from our proposed TASE-Net using the noise-free RF signals emitted by the eight UAVs in the DroneRFa dataset [34].

$\rho = 0.95$ is the appropriate choice accordingly since it leads to the best results.

2) *FFT Window Size and Sample Size for CPP Extraction*: Now we vary both FFT window size N' and sample size N to evaluate the corresponding accuracies (P_{cc} 's) and run-times to investigate the corresponding tradeoff. Other simulation settings remain unchanged as before. Fig. 16 shows the pertinent results. Because the accuracy (P_{cc}) is the most important performance measure, we can find that both $(N' = 512, N = 32768)$ and $(N' = 256, N = 16384)$ lead to

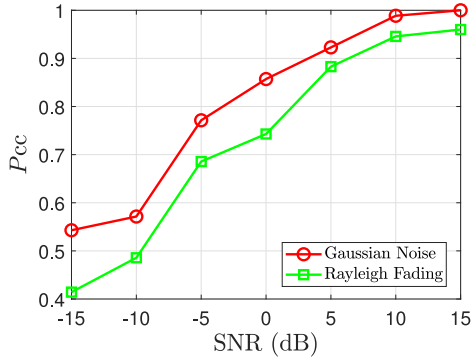


Fig. 17. P_{cc} 's (open-set classification) versus SNR resulting from our proposed TASE-Net subject to $\psi = 12.5\%$ in the five-shot training scenario. The channel models are assumed to be AWGN channel and Rayleigh-fading channel.

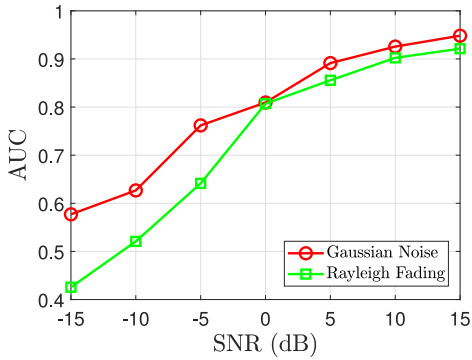


Fig. 18. AUCs for unknown-class rejection versus SNR resulting from our proposed TASE-Net subject to $\psi = 12.5\%$ in the five-shot training scenario. The channel models are assumed to be AWGN channel and Rayleigh-fading channel.

the perfect accuracy $P_{cc} = 100\%$ according to Fig. 16. As ($N' = 256$, $N = 16384$) requires much less computational complexity than ($N' = 512$, $N = 32768$), we choose ($N' = 256$, $N = 16384$) to seek the best tradeoff between the accuracy and the required computational complexity in this work.

D. Robustness Analysis

We have also investigated the robustness of our proposed novel TASE-Net in the presence of *burst interference*, *channel fading*, or *narrowband interference*. The simulation settings still remain the same as before. Figs. 17 and 18 delineate the corresponding P_{cc} 's and AUCs resulting from our proposed new TASE-Net for open-set few-shot UAV recognition subject to AWGN and Rayleigh fading channels. According to Figs. 17 and 18, the TASE-Net performance would degrade in the presence of Rayleigh fading. On the other hand, two typical kinds of interference, namely, burst interference and narrowband interference, are considered here. A rectangular-pulse burst spanning over a hundred sample intervals is generated by computer. Meanwhile, a single-frequency (10 MHz) sinusoidal sequence is also produced by computer as a narrowband interference. The signal-to-interference ratio (SIR) of the interfered UAV RF signal (by both burst and narrowband interferences) ranges from -15 dB to 15 dB while

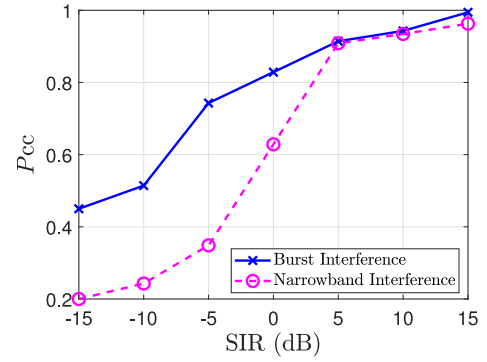


Fig. 19. P_{cc} 's (open-set classification) versus SIR resulting from our proposed TASE-Net subject to $\psi = 12.5\%$ in the five-shot training scenario. Here, the UAV RF signals suffer from burst interference and narrowband interference.

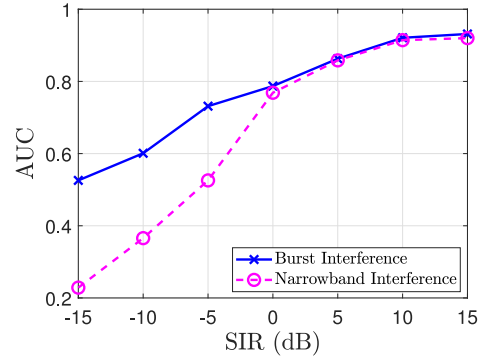


Fig. 20. AUCs for unknown-class rejection versus SIR resulting from our proposed TASE-Net subject to $\psi = 12.5\%$ in the five-shot training scenario. Here, the UAV RF signals suffer from burst interference and narrowband interference.

other simulation parameters remain unchanged as before. Figs. 19 and 20 show the corresponding P_{cc} 's and AUCs resulting from our proposed new TASE-Net for open-set few-shot UAV recognition in the presence of the aforementioned burst and narrowband interferences. According to Figs. 19 and 20, our proposed new TASE-Net would be more sensitive to narrowband interference especially when the SIR condition is poor (i.e., $SIR < 0$ dB).

E. Computational-Complexity Analysis

At last but not least, we would like to present the computational-complexity analysis of the aforementioned schemes. In our simulations, all methods in comparison are carried out using the workstation configured with an Intel Xeon GOLD 5222 CPU, 64-GB RAM, and a double NVIDIA GeForce RTX 2080TI GPU. The model complexities (in terms of number of trainable parameters) and the run-times (for training and testing⁴) of our proposed TASE-Net and three other existing DL models in [18] and [22] are summarized by Table II for the five-shot scenario. Note that CPP features are adopted here. According to Table II, our proposed TASE-Net requires the second least complex DL architecture, the

⁴The "testing time" is referred to as the inference time required to recognize a single CPP tensor, which excludes the time required for UAV signal acquisition and feature extraction.

TABLE II
COMPARISON OF MODEL-COMPLEXITIES AND RUN-TIMES
REQUIRED BY FOUR MODELS (FIVE-SHOT SCENARIO)

Scheme	Number of Trainable Parameters	Training Time (hour)	Testing Time (sec)
CPP+MSCL	3.295×10^7	1.323	0.975
CPP+modified-MSCL	3.401×10^7	1.350	0.998
CPP+OpenMax	1.055×10^7	0.532	0.273
CPP+TASE-Net	1.329×10^7	0.396	0.456

shortest training run-time, and the second shortest testing run-time among all. As the training run-time is most important, our proposed new TASE-Net is the best in this aspect. Thus, our proposed new UAV RF signal recognition approach appears to be very memory- and computationally-efficient in comparison with other existing DL models.

IV. CONCLUSION

Open-set few-shot UAV recognition is quite challenging as more and more UAVs are employed nowadays and it is impossible to train all UAV signals in reality. In this work, we propose a novel robust open-set few-shot UAV recognition approach by segmenting a received UAV RF signal into frames and then applying the ST-ESER to select the predominant signal frames, which will further lead to CPP tensors. Meanwhile, we design a novel DL model, a.k.a. TASE-Net, for open-set few-shot UAV identification using CPP features. Monte Carlo simulation results justify the superiority of our proposed new open-set few-shot UAV recognition approach to other existing DL models. Moreover, our proposed TASE-Net is rather memory- and computationally efficient in comparison with other existing DL models.

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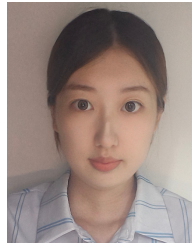
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